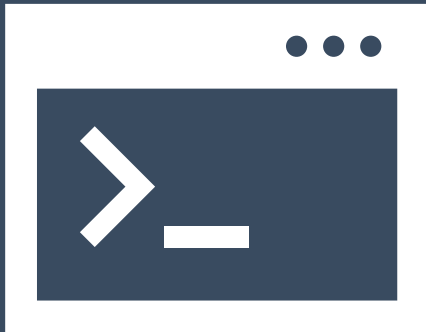


Investigating chromatin biology using ChIP-seq and CUT&RUN

<https://tinyurl.com/hbc-chromatin>

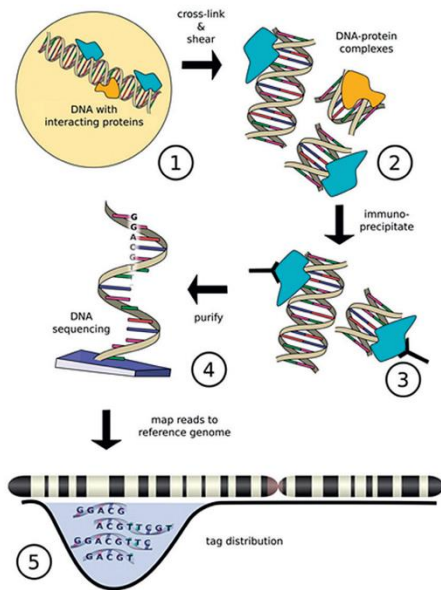


Harvard Chan Bioinformatics Core
in collaboration with
HMS Research Computing



Genomic Methods for Profiling Chromatin

ChIP-seq



CUT&RUN

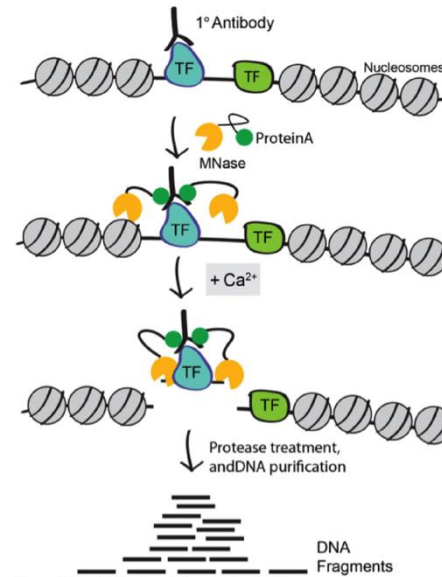
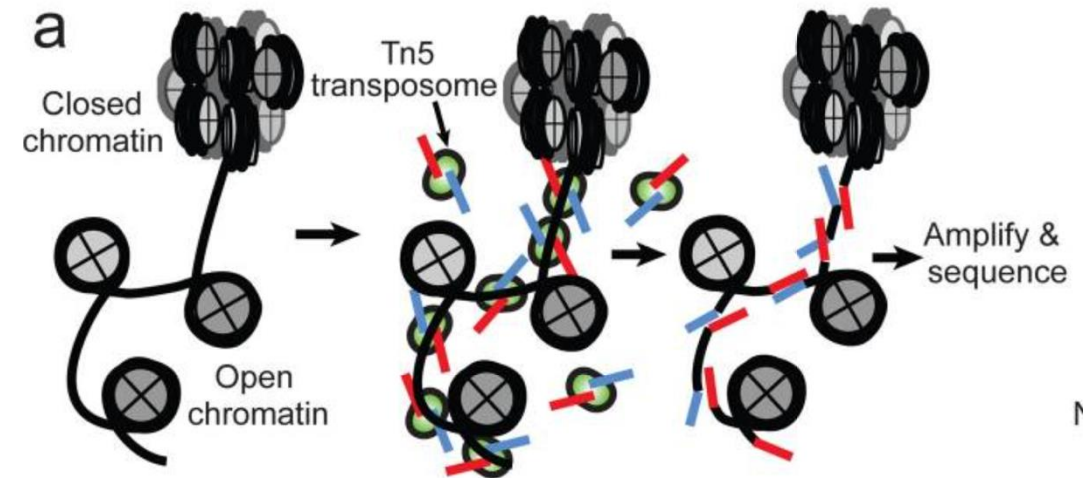
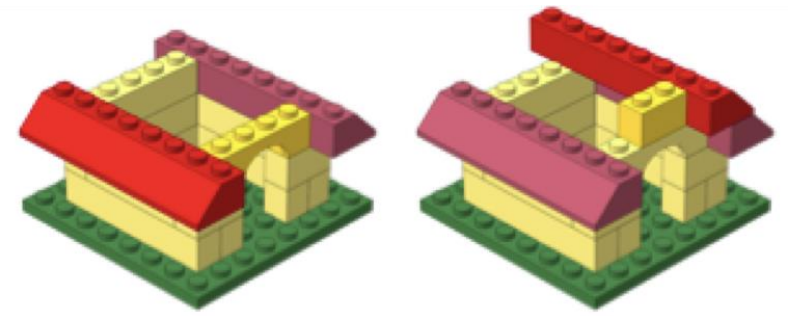


Figure 1. CUT&RUN schematic (see text for details).

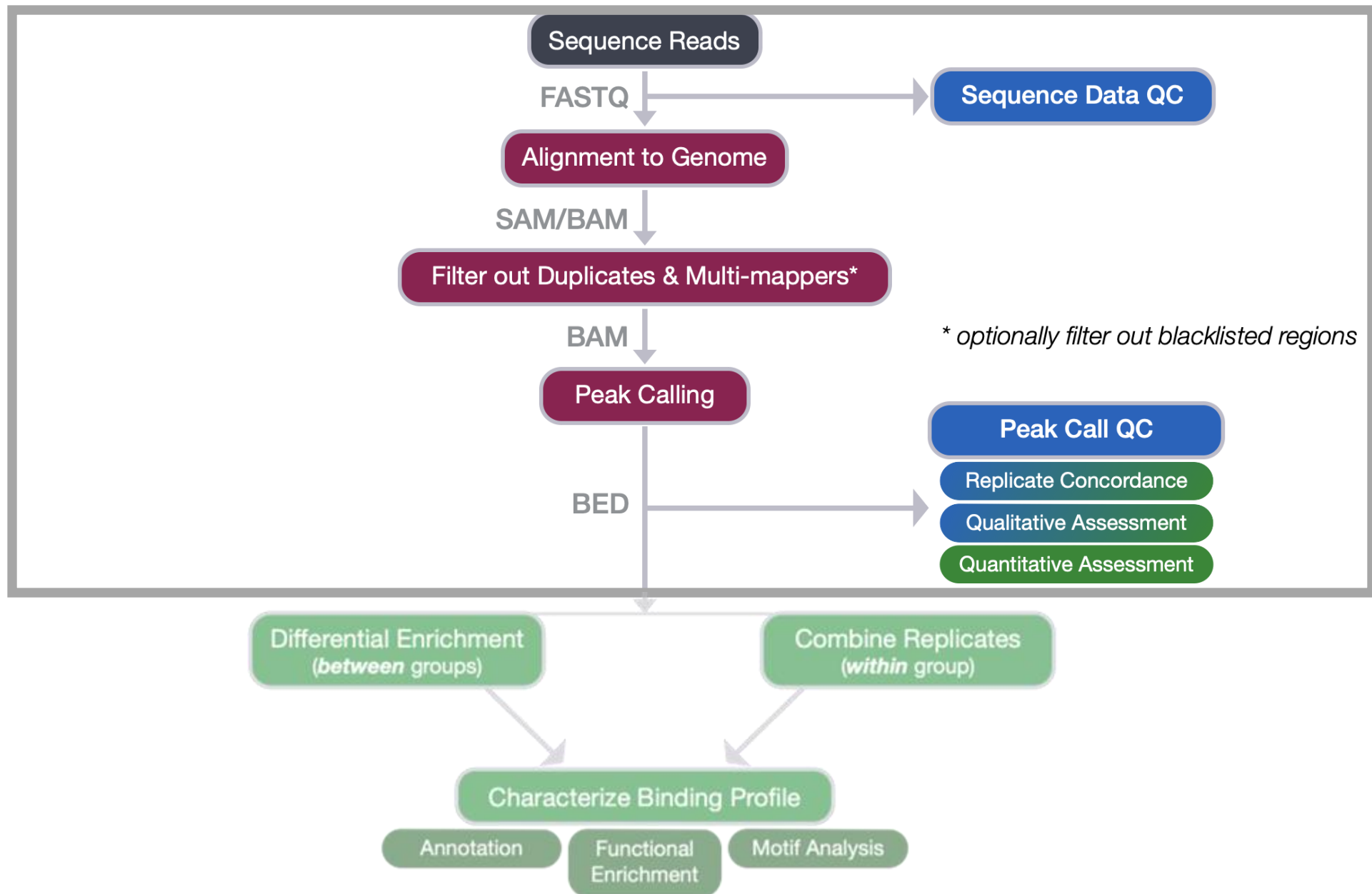
ATAC-seq



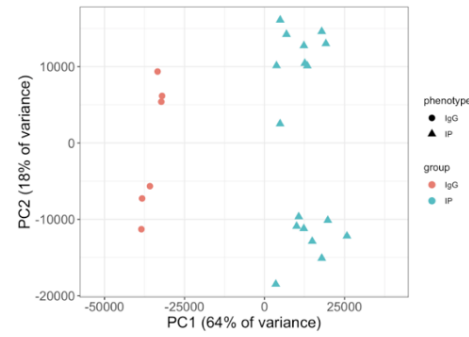
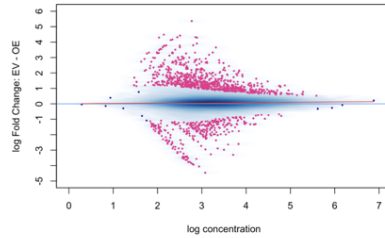
Learning Objectives



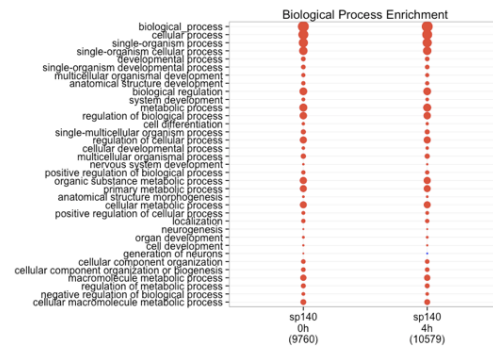
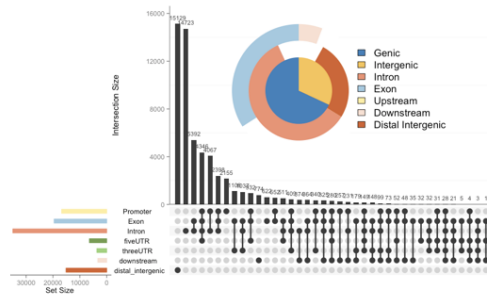
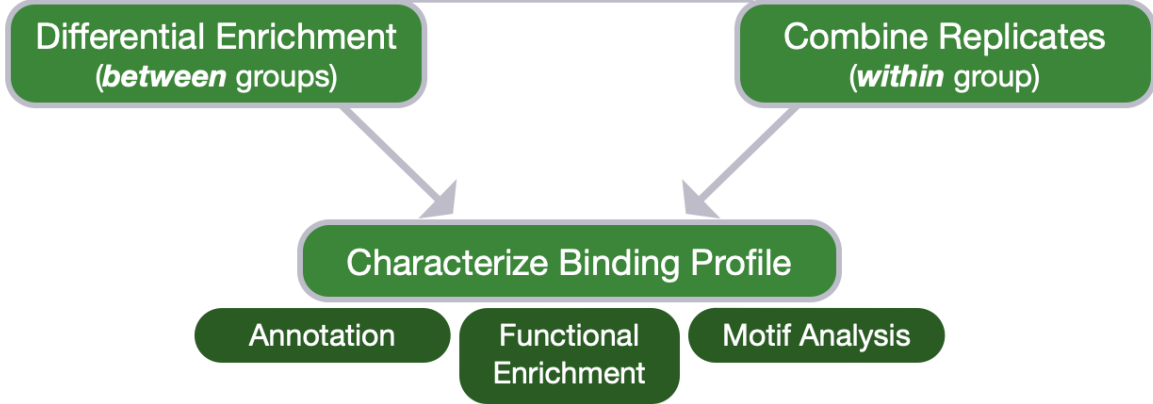
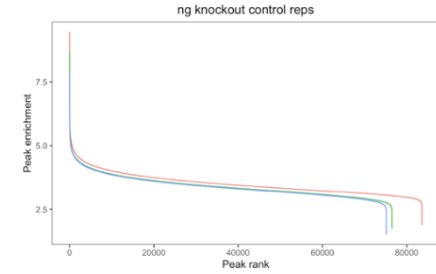
- ❖ Describe important considerations for setting up a successful ChIP-seq, CUT&RUN or ATAC-seq experiment
- ❖ Describe the steps in an ChIP-seq analysis workflow (from sequence data to peak calls) and contrast any differences for CUT&RUN and ATAC-seq analyses
- ❖ Learn how to handle various file formats encountered when analyzing ChIP-seq and related data
- ❖ Implement shell scripts on a high-performance compute cluster to perform the above steps



Boxes in green represent parts of the workflow that will not be covered in this



Peak enrichment vs. Peak rank



For further information on how to interpret these results or to get a copy of the MEME software please access <http://meme.nbcr.net>.

If you use DREME in your research please cite the following paper:
Timothy L. Bailey, "DREME: Motif discovery in transcription factor ChIP-seq data", *Bioinformatics*, 27(12):1653-1659, 2011. [\[full text\]](#)

[DISCOVERED MOTIFS](#) | [INPUTS & SETTINGS](#) | [PROGRAM INFORMATION](#)

DISCOVERED MOTIFS

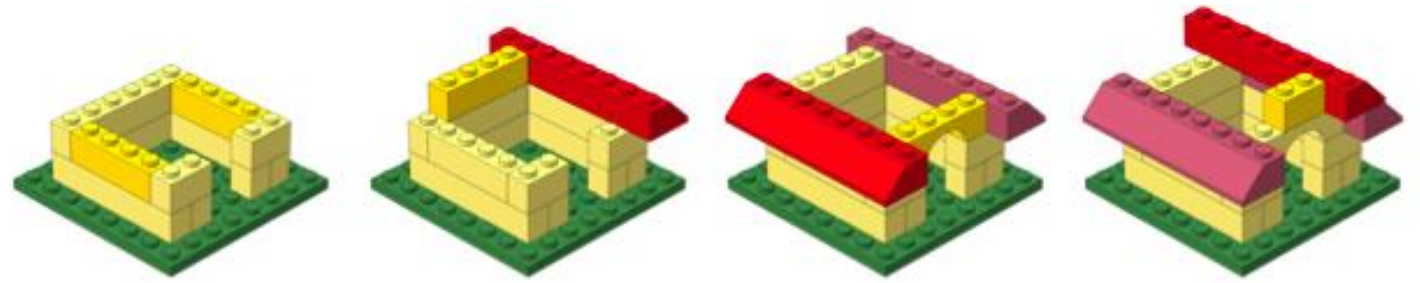
Motif	Logo	RC Logo	E-value	Unersered E-value	More	Submit/Download
1. CYWTTGTB			4.2e-299	4.2e-299	↓	→
2. ATGBAAAT			8.4e-179	1.1e-179	↓	→
3. CCMCDCCC			1.3e-130	1.1e-131	↓	→

Downstream of Peak Calls

Exit survey

<http://tinyurl.com/hbc-chromatin-bio-exitsurvey>

Keep building!



Topic	Pre-requisites	Date	Time	Registration
Coding with Others: Managing Conflicts on GitHub	"Track Changes" for your code: An Introduction to Git and GitHub	5/21/25	1 - 4 pm	Register now!
Statistics for Computational Biology Projects	None	6/18/25	1 - 4 pm	Register now!
Deeper differential expression analysis with shrinkage correction	Foundations in R	7/16/25	1 - 4 pm	Register now!

Get an O2 account!

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Search IT services and kn

Q

O2 Cluster Account

Description

What does it do?

Get an account on O2, the HMS High Performance Compute cluster

Available To

Who is eligible?

Any HMS or HMS-affiliated researcher who has an HMS eCommons account.

Cost

What does it cost?

There is no cost to labs whose PI has a primary or secondary faculty appointment in an HMS Quad department.

If the PI of your lab does not have a primary or secondary faculty appointment in an HMS Quad department, cluster usage will be charged for beginning later in 2021. Please see the following page for current details about rates: <https://it.hms.harvard.edu/rc/core/rates>.

Support

Please fill out the [online help request form](#), or email rchelp@hms.harvard.edu.

How to Access

How do I get it?

Two-factor authentication is required to request an account on O2, as well as for O2 logins once your account is created. Harvard University uses a mobile app called Duo that makes the process quick and easy. Even if you already use Duo for HarvardKey, you will still need to setup a Duo profile for HMS.

- [Setup HMS two-factor Authentication \(HMS Duo Mobile\)](#)
- [Reactivate or Reconnect HMS two-factor Authentication \(HMS Duo Mobile\)](#)

Once you have Duo set up, Click the "Get Service" link to login and complete the request form.

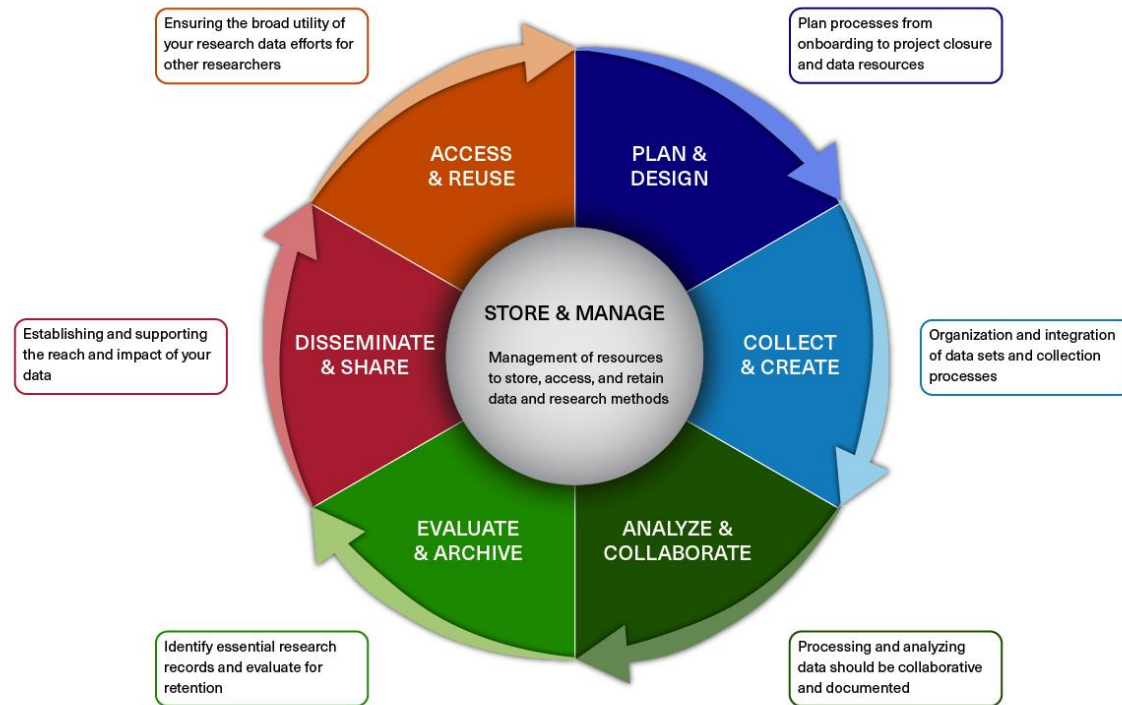
Get this service

Learn more

Don't see what you're looking for?

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Spring 2025 Data Lifecycle Training

Plan & Design

February 11 

Data Management
Offboarding for
Research Projects

March 19  

A Guide to Efficient
Research Practices

March 26  

Tips and Tricks for Writing
an Actionable Data
Management Plan

May 21 

Research Data
Stewardship Basics

Collect & Analyze

January 29 

Data Literacy:
Introduction to GIS

February 19 

Foundations in R

March 19 

Reproducible Research
using RMarkdown

April 16 

An Introduction to Git
and GitHub

May 21 

Managing Conflicts
on GitHub

Store & Evaluate

March 10 

Introduction to the
General Records Schedule

April 7 

Managing Paper Records:
Off-Site Records Storage

April 21 

Managing Electronic
Records: Shared Drives
and Emails

Share & Publish

February 27 

Research Management:
Open Access Publishing

April 9  

Research Management:
Closing Out Your Research

April 23 

Data Sharing with
Harvard Dataverse

 In-person

 Virtual



Learn More & Register
bit.ly/rdmwg-calendar



<https://datamanagement.hms.harvard.edu/about/news-events/rdmwg-calendar>

Better RDM practice benefits you

- ❖ **HMS Data Management LMA**

- ❖ **Webpage:** <https://datamanagement.hms.harvard.edu>

- ❖ **Sign up for quarterly email updates**

- ❖ **Harvard-wide Research data Management**

- ❖ <https://researchdatamanagement.harvard.edu/>

Join us for HBC Community Breakfast!

- ❖ An opportunity to get to know others in the community
- ❖ Free food and beverages
- ❖ Great conversations



TBD
9:00 to 10:30am

More Info:

<http://bioinformatics.sph.harvard.edu/breakfast/>

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu
- ❖ *O2 (HMS-RC):* rchelp@hms.harvard.edu

Talk to us early!

Involvement in study design to optimize experiments



Thanks!

- ❖ Dr. Ming “Tommy” Tang
- ❖ Kathleen Chappell and Andy Bergman from HMS-RC
- ❖ Data Carpentry

These materials have been developed by members of the teaching team at the Harvard Chan Bioinformatics Core (HBC). These are open access materials distributed under the terms of the Creative Commons Attribution license (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

More Information

- ❖ *HBC training materials: <https://hbctraining.github.io/main>*
- ❖ *HBC website: <http://bioinformatics.sph.harvard.edu>*
- ❖ *O2 Wiki (HMS-RC): <https://wiki.rc.hms.harvard.edu/display/O2>*